



Caringbah High School

2017

Trial HSC Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen (Black pen is preferred)
- Board-approved calculators may be used
- A Board-approved reference sheet is provided for this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 2 – 5

10 marks

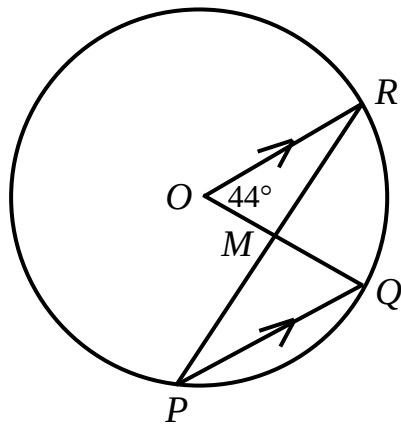
- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6 – 12

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

4)



NOT TO SCALE

O is the centre of the circle.

OR is parallel to PQ .

$\angle ROM = 44^\circ$.

What is the size of $\angle RMQ$?

A) 22°

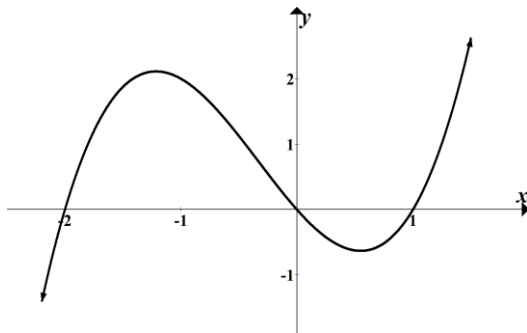
B) 44°

C) 66°

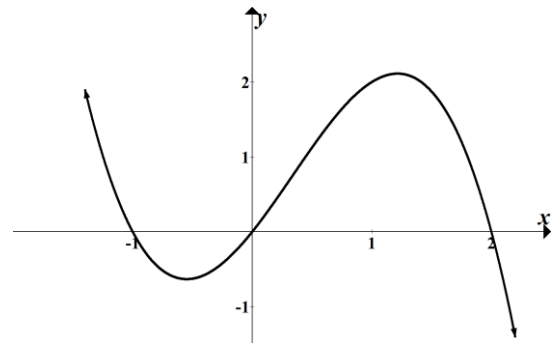
D) 88°

5) Which of the following could be the graph of $y = 2x - x^2 - x^3$?

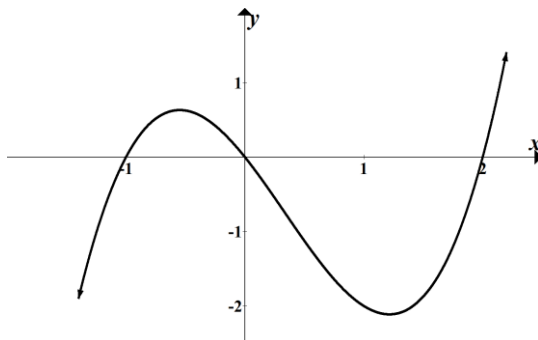
A)



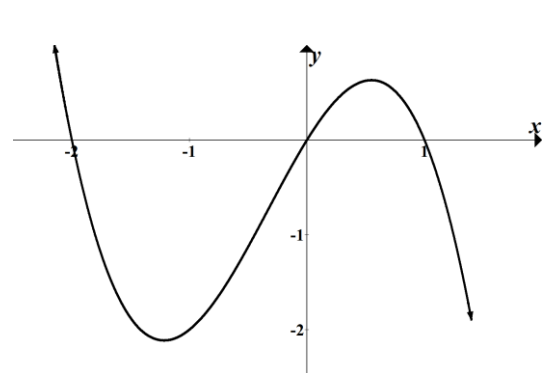
B)



C)



D)



- 6) What are the asymptotes of the curve defined by the

parametric equations $x = \frac{1}{t}, y = \frac{t}{t+1}$?

A) $x = -1, y = 0$

B) $x = 1, y = -1$

C) $x = 0$ only

D) $x = -1$ only

- 7) To the nearest degree, what is the acute angle between the lines

$2x + y = 5$ and $x - 3y = 1$?

A) 18°

B) 45°

C) 72°

D) 82°

- 8) What is the value of k such that $\int_0^{2k} \frac{1}{\sqrt{3-x^2}} dx = \frac{\pi}{3}$?

A) $\frac{\sqrt{3}}{4}$

B) $\frac{\sqrt{3}}{2}$

C) $\frac{3}{4}$

D) $\frac{3}{2}$

- 9) Consider the function $f(x) = x^3 - 12x$.

What is the largest possible domain containing the origin for which $f(x)$ has an inverse function $f^{-1}(x)$?

A) $-2 < x < 2$

B) $-2 \leq x \leq 2$

C) $-2\sqrt{3} < x < 2\sqrt{3}$

D) $-2\sqrt{3} \leq x \leq 2\sqrt{3}$

- 10) The acceleration of a particle is defined in terms of its position by $\ddot{x} = 2x^3 + 4x$.

The particle is initially 2 metres to the right of the origin, travelling at 6ms^{-1} .

What is the minimum speed of the particle?

A) 36ms^{-1}

B) 20ms^{-1}

C) 16ms^{-1}

D) 6ms^{-1}

END OF MULTIPLE CHOICE QUESTIONS

Section II**60 marks****Attempt all questions 11–14****Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW booklet.**Marks**

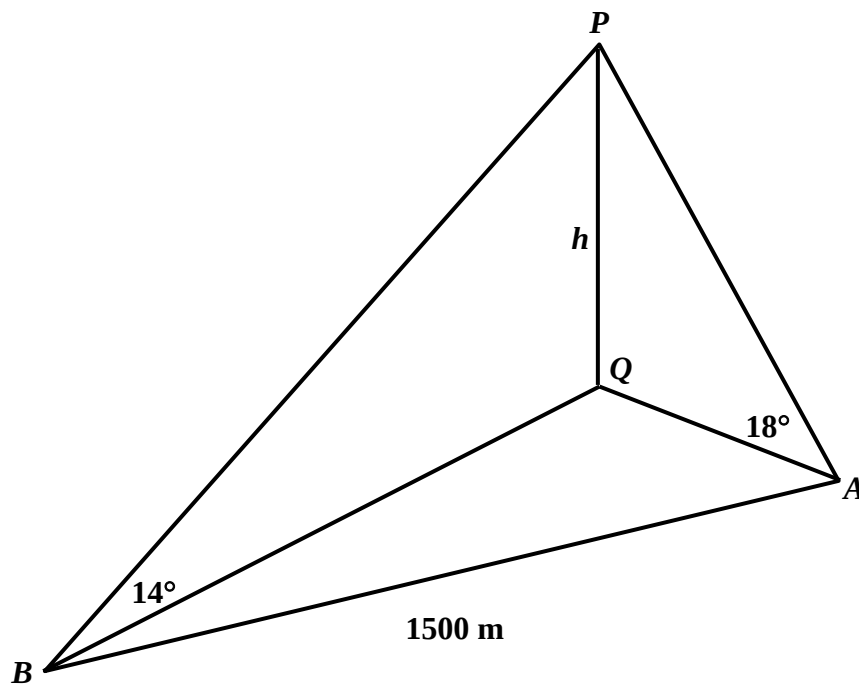
- a) Find $\frac{d}{dx}[\log_e(\ln x)]$. **1**
- b) Show that $\lim_{\theta \rightarrow 0} \frac{5\theta}{\tan \frac{\theta}{2}} = 10$. **2**
- c) If $y = be^{ax}$ find $\frac{d^2y}{dx^2}$ in terms of y . **2**
- d) Find $\int \cos^2 2x \, dx$. **2**
- e) Prove $\frac{2\cos\theta}{\operatorname{cosec}\theta - 2\sin\theta} = \tan 2\theta$. **2**
- f) Solve $\frac{x^2 - 3}{2x} \geq 0$. **3**
- g) Solve $e^t - e^{-t} = 2$, expressing the answer in simplest exact form. **3**

Question 12 (15 marks) Start a NEW booklet.**Marks**

- a) i) Express $\sin \theta - \sqrt{3} \cos \theta$ in the form $A \sin(\theta - \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$. **2**
- ii) Hence, or otherwise, find the minimum value of $\sin \theta - \sqrt{3} \cos \theta$. **1**

- b) Use the substitution $u = x^4 - 1$ to evaluate $\int_0^1 \frac{x^3}{1 + x^8} dx$. **3**

- c) The angle of elevation of a tower PQ of height h metres at a point A due east of it is 18° . From another point B , due south of the tower the angle of elevation is 14° . The points A and B are 1500 metres apart on level ground.



- i) Show that $BQ = h \tan 76^\circ$. **1**
- ii) Find the height h of the tower to the nearest metre. **2**

Question 12 continues on page 8

Question 12 (continued)

Marks

- d) i) Show that $P(x) = 3\sin^{-1}\left(\frac{x}{2}\right) - 2x$ is an odd function. **1**
- ii) Carefully sketch $y = 3\sin^{-1}\left(\frac{x}{2}\right)$ and $y = 2x$ on the same number plane. **2**
- iii) It is known that a close approximation to a root of $P(x) = 0$ is $x_1 = 1.9$. **2**
- Use one application of Newton's method to find a better approximation x_2 . Give your answer correct to 3 decimal places.
- iv) Find the sum of the roots of $P(x) = 0$. **1**

End of Question 12

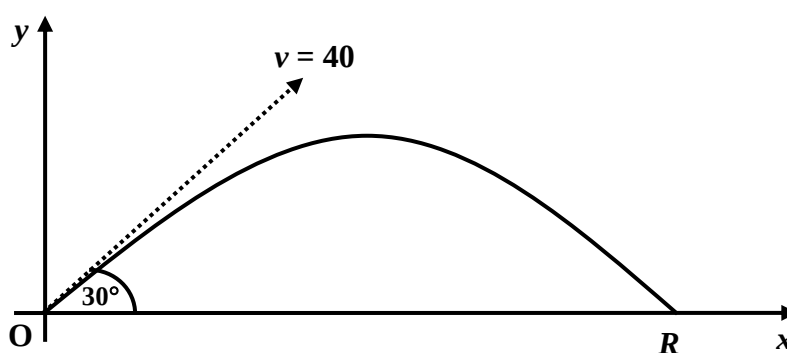
Question 13 (15 marks) Start a NEW booklet.**Marks**

- a) Use mathematical induction to prove that $13 \times 6^n + 2$ is divisible by 5 for all integers $n \geq 1$.

3

- b) A particle is projected with an initial velocity of 40 m s^{-1} at an angle of 30° to the horizontal. The equations of motion are given by

$$x = 20\sqrt{3}t, \quad y = 20t - 5t^2. \quad (\text{Do NOT prove this.})$$



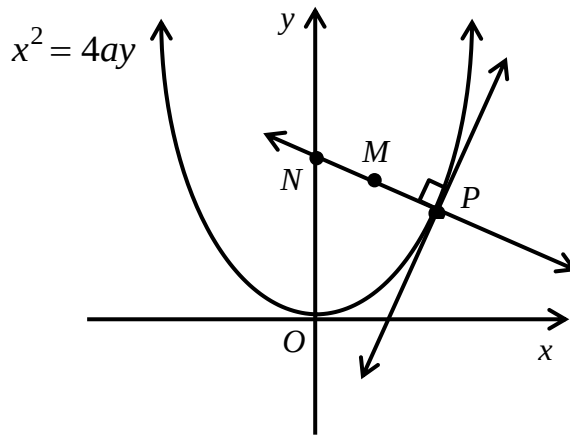
- i) Find the maximum height of the particle.
- ii) Find the range (R) of the particle.
- iii) Determine the cartesian equation of the particles flight path.

2**1****1****Question 13 continues on page 10**

Question 13 (continued)

Marks

- c) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$. The normal to the parabola at P cuts the y -axis at N . M is the midpoint of PN .

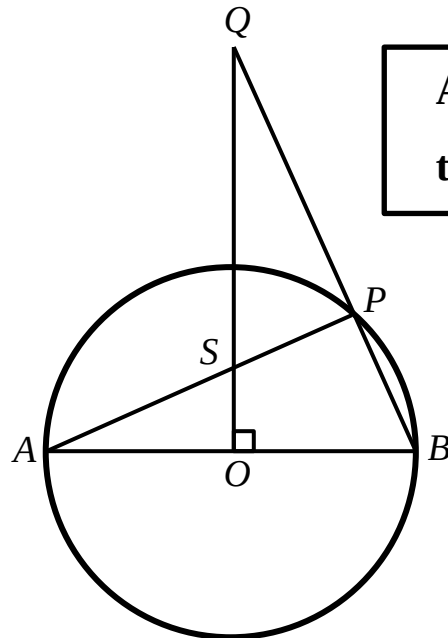


- i) Show that N has coordinates $(0, ap^2 + 2a)$. 1
- ii) Show that the locus of M as P moves on the parabola $x^2 = 4ay$ is another parabola and state its focal length. 3
- d) The velocity $v \text{ m s}^{-1}$ of a particle moving in simple harmonic motion along the x -axis is given by $v^2 = 5 + 4x - x^2$.
- i) Between which two points is the particle oscillating? 1
- ii) What is the amplitude of the motion? 1
- iii) Find the acceleration of the particle in terms of x . 1
- iv) Find the period of oscillation. 1

End of Question 13

Question 14 (15 marks) Start a NEW booklet.**Marks**

- a) The circle centred at O has a diameter AB . BPQ and ASP are straight lines. Also, the line OSQ is perpendicular to AOB .



**Answer this question on
the page provided**

- | | | |
|-----|---|---|
| i) | Show that $AOPQ$ is a cyclic quadrilateral. | 2 |
| ii) | Show that $\angle AQO = \angle OBS$. | 2 |
-
- b) The roots α, β and γ of the equation $2x^3 + 9x^2 - 27x - 54 = 0$ are consecutive terms of a geometric sequence.
- | | | |
|------|---|---|
| i) | Show that $\beta^2 = \alpha\gamma$. | 1 |
| ii) | Find the value of $\alpha\beta\gamma$. | 1 |
| iii) | Find the roots α, β and γ . | 3 |

Question 14 continues on page 12

Question 14 (continued)

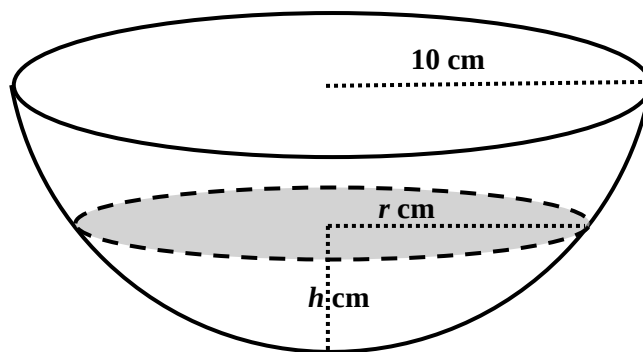
Marks

- c) Let a hemispherical bowl of radius 10 cm contain water to a depth of h cm.

The volume of water in the bowl in terms of its depth h cm is given by

$$V = \frac{1}{3}\pi h^2(30 - h) \text{ cm}^3.$$

Water is poured into the bowl at a rate of $2 \text{ cm}^3 \text{ s}^{-1}$.



- i) If the radius of the water surface is r cm, show that $r = \sqrt{20h - h^2}$. **1**

When the depth of the water is 4 cm, find in terms of π :

- ii) the rate of change of the water level. **2**
- iii) the rate of change of the radius of the water surface. **3**

End of paper

Multiple Choice Section:

1.B 2.A 3.C 4.C 5.D
 6.A 7.D 8.C 9.B 10.D

Question 1.

The remainder is given by $P(2)$

$$= 2(8) + 1 - 10(4) + 4 = -20 \quad \text{-----} \boxed{B}$$

Question 2.

$$x = \frac{2 \times -2 + 3 \times 8}{2 + 3} = 4 \quad \text{-----} \boxed{A}$$

Question 3.

$$\begin{aligned} \tan\left(\frac{\pi}{4} - x\right) &= \frac{\tan\frac{\pi}{4} - \tan x}{1 + \tan\frac{\pi}{4} \cdot \tan x} \\ &= \frac{1 - \tan x}{1 + \tan x} = \frac{1 - \frac{\sin x}{\cos x}}{1 + \frac{\sin x}{\cos x}} \\ &= \frac{\cos x - \sin x}{\cos x + \sin x} \quad \text{-----} \boxed{C} \end{aligned}$$

Question 4.

$$\angle RPQ = 22^\circ \quad \{\angle \text{ at centre is twice } \angle \text{ at circumference}\}$$

$$\angle ORM = 22^\circ \quad \{\text{alt } \angle \text{'s, OR} \parallel \text{PQ}\}$$

$$\angle RMQ = 66^\circ \quad \{\text{ext } \angle \text{ of triangle}\} \quad \text{-----} \boxed{C}$$

Question 5.

Using roots ($x = 0, 1, -2$) and negative cubic

$$\text{-----} \boxed{D}$$

Question 6.

$$x = \frac{1}{t}, y = \frac{t}{t+1} \rightarrow t = \frac{1}{x}$$

$$\therefore y = \frac{\frac{1}{x}}{\frac{1}{x} + 1} = \frac{1}{1+x}$$

$$\therefore x \neq -1 \text{ and } y \neq 0 \quad \text{-----} \boxed{A}$$

Question 7.

$$2x + y = 5 \rightarrow m_1 = -2$$

$$x - 3y = 1 \rightarrow m_2 = \frac{1}{3}$$

$$\therefore \tan \theta = \left| \frac{-2 - \frac{1}{3}}{1 + (-2) \times \frac{1}{3}} \right| = 7$$

$$\therefore \theta = 82^\circ \quad (\text{NOTE: accept } 81^\circ 52') \quad \text{-----} \boxed{D}$$

Question 8.

$$\int_0^{2k} \frac{1}{\sqrt{3-x^2}} dx = \frac{\pi}{3}$$

$$\therefore \left[\sin^{-1}\left(\frac{x}{\sqrt{3}}\right) \right]_0^{2k} = \frac{\pi}{3}$$

$$\therefore \sin^{-1}\left(\frac{2k}{\sqrt{3}}\right) - \sin^{-1}(0) = \frac{\pi}{3}$$

$$\therefore \frac{2k}{\sqrt{3}} = \sin \frac{\pi}{3} \rightarrow \frac{2k}{\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$\therefore k = \frac{3}{4} \quad \text{-----} \boxed{C}$$

Question 9.

$$f(x) = x^3 - 12x \rightarrow f'(x) = 3x^2 - 12$$

$$f'(x) = 0 \text{ for stationary points}$$

$$\therefore 3(x-2)(x+2) = 0 \rightarrow x = \pm 2$$

$$\text{hence } -2 \leq x \leq 2 \quad \text{-----} \boxed{B}$$

Question 10.

$$v^2 = 2 \int 2x^3 + 4x \, dx$$

$$\therefore v^2 = x^4 + 4x^2 + c$$

$$\text{when } x=2, v=6$$

$$\therefore 36 = 32 + c \rightarrow c = 4$$

$$\therefore v^2 = x^4 + 4x^2 + 4 = (x^2 + 2)^2$$

$$\text{hence minimum speed when } x = 2$$

$$\therefore v^2 = (4 + 2)^2 \rightarrow v = 6 \quad \text{-----} \boxed{D}$$

Question 11.

$$\text{a) } \frac{d}{dx} [\log_e (\ln x)] = \frac{\frac{1}{x}}{\ln x} = \frac{1}{x \ln x}$$

$$\begin{aligned} \text{b) } \lim_{\theta \rightarrow 0} \frac{5\theta}{\tan \frac{\theta}{2}} &= 5 \lim_{\theta \rightarrow 0} \frac{\frac{\theta}{2}}{\frac{1}{2} \tan \frac{\theta}{2}} \\ &= \frac{5}{\frac{1}{2}} \times 1 = 10 \end{aligned}$$

$$\text{c) } y = be^{ax} \rightarrow y' = abe^{ax} \rightarrow y'' = a^2 (be^{ax})$$

$$\therefore y'' = a^2 y$$

$$\text{d) } \cos 2x = 2\cos^2 x - 1 \rightarrow \cos^2 2x = \frac{1}{2}(\cos 4x + 1)$$

$$\therefore I = \frac{1}{2} \int \cos 4x + 1 \, dx$$

$$= \frac{1}{2} \left(\frac{1}{4} \sin 4x + x \right) + c$$

$$= \frac{1}{8} \sin 4x + \frac{1}{2} x + c$$

$$\text{e) } LHS = \frac{2\cos\theta}{\frac{1}{\sin\theta} - 2\sin\theta}$$

$$= \frac{2\cos\theta}{\frac{1-2\sin^2\theta}{\sin\theta}} = \frac{2\sin\theta\cos\theta}{1-2\sin^2\theta}$$

$$= \frac{\sin 2\theta}{\cos 2\theta} = \tan 2\theta = RHS$$

$$\text{f) CV1 and CV2: } x^2 - 3 = 0 \rightarrow x = \pm\sqrt{3}$$

$$\text{CV3: } x = 0$$

$$\text{On testing } -\sqrt{3} \leq x < 0 \text{ and } x \geq \sqrt{3}, \text{ note } x \neq 0$$

$$\text{g) } e^t - \frac{1}{e^t} = 2 \rightarrow (e^t)^2 - 2e^t - 1 = 0$$

$$\therefore e^t = \frac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2}$$

$$\therefore e^t = 1 + \sqrt{2} \text{ or } e^t = 1 - \sqrt{2}$$

$$\therefore t = \ln(1 + \sqrt{2}) \text{ or } t = \ln(1 - \sqrt{2})$$

$$\text{but since } \ln(1 - \sqrt{2}) \text{ does not exist}$$

$$\text{then } t = \ln(1 + \sqrt{2}) \text{ only.}$$

Question 12.

$$\begin{aligned} \text{a)i) Let } \sin\theta - \sqrt{3}\cos\theta &= A\sin(\theta - \alpha) \\ &= A\cos\alpha\sin\theta - A\sin\alpha\cos\theta \end{aligned}$$

Equating coefficients of $\cos x$ and $\sin x$ gives:

$$A\cos\alpha = 1 \text{ ----} \boxed{1} \text{ and } A\sin\alpha = \sqrt{3} \text{ ----} \boxed{2}$$

$$\boxed{1}^2 + \boxed{2}^2 \rightarrow A^2(\cos^2\alpha + \sin^2\alpha) = 4$$

$$\therefore A^2 = 4 \rightarrow A = 2$$

Using [1] with $A=2 \rightarrow \cos \alpha = \frac{1}{2}$

$$\therefore \alpha = \frac{\pi}{3}$$

$$\therefore \sin \theta - \sqrt{3} \cos \theta = 2 \sin \left(\theta - \frac{\pi}{3} \right)$$

ii) minimum value $= -2$ since $-1 \leq \sin A \leq 1$

$$\text{b) } \int_0^1 \frac{x^3}{1+x^8} dx \quad u = x^4 - 1 \rightarrow du = 4x^3 dx$$

$$x^4 = u + 1 \rightarrow x^8 = (u + 1)^2$$

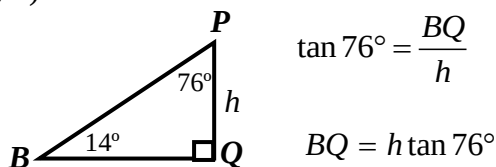
when $x=0, u=-1; x=1, u=0$

$$\therefore I = \frac{1}{4} \int_{-1}^0 \frac{1}{1+(u+1)^2} du$$

$$= \frac{1}{4} \left[\tan^{-1}(u+1) \right]_{-1}^0$$

$$= \frac{1}{4} \left(\tan^{-1} 1 - \tan^{-1} 0 \right) = \frac{\pi}{16}$$

c) i)



ii) Similarly using $\triangle PQA$: $AQ = h \tan 72^\circ$

$$\text{In } \triangle BQA: BQ^2 + AQ^2 = 1500^2$$

$$\therefore h^2 \tan^2 76^\circ + h^2 \tan^2 72^\circ = 1500^2$$

$$\therefore h^2 (\tan^2 76^\circ + \tan^2 72^\circ) = 1500^2$$

$$\therefore h^2 = \frac{1500^2}{\tan^2 76^\circ + \tan^2 72^\circ} \rightarrow h = 297 \text{ m (nearest metre)}$$

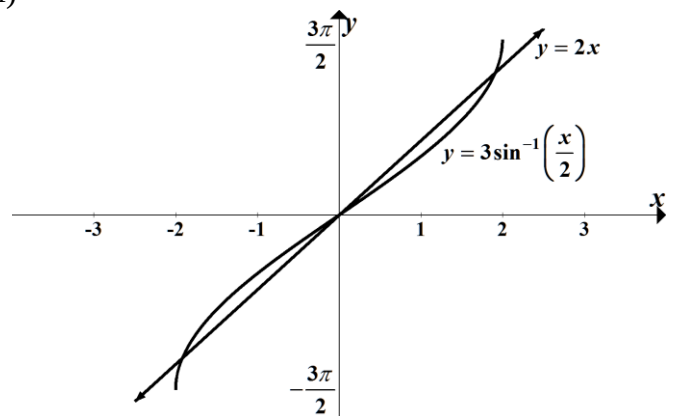
d) i) Function will be odd if $P(x) = -P(-x)$

$$-P(-x) = - \left[3 \sin^{-1} \left(-\frac{x}{2} \right) - 2(-x) \right]$$

$$= - \left[-3 \sin^{-1} \left(\frac{x}{2} \right) + 2(x) \right]$$

$$= 3 \sin^{-1} \left(\frac{x}{2} \right) - 2(x) = P(x)$$

ii)



iii)

$$P(x) = 3 \sin^{-1} \left(\frac{x}{2} \right) - 2x \rightarrow P(1.9) = -0.04029$$

$$P'(x) = \frac{3}{\sqrt{4-x^2}} - 2 \rightarrow P'(1.9) = 2.8038$$

$$\therefore x_2 = 1.9 - \frac{-0.04029}{2.8038} \approx 1.914(3dp)$$

iv) zero (odd function has roots $-\alpha, 0, \alpha$)

Question 13.

a) $13 \times 6^n + 2$

$$\text{When } n=1: 13 \times 6^1 + 2 = 13 \times 6^1 + 2$$

$$= 80 = 5 \times 16$$

hence true for $n=1$

Assume true for $n=k$ and let

$$13 \times 6^k + 2 = 5M \quad (M \text{ an integer}) \text{-----**}$$

Prove true for $n=k+1$

$$\text{Hence } 13 \times 6^{k+1} + 2 = 13 \times 6 \times 6^k + 2$$

$$\begin{aligned} &= 6 \times [13 \times 6^k] + 2 \\ &= 6 \times [5M - 2] + 2 \text{ using **} \\ &= 30M - 10 \\ &= 5(6M - 2) \end{aligned}$$

$\therefore$ true for $n = k + 1$

Hence by the principle of mathematical induction,
the result is true for all $n \geq 1$.

b) i) $x = 20\sqrt{3}t, \quad y = 20t - 5t^2$

Maximum height when $\dot{y} = 0$

$$\therefore \dot{y} = 20 - 10t$$

$$\therefore 20 - 10t = 0 \rightarrow t = 2$$

$$\text{when } t = 2, \quad y = 20 \times 2 - 5 \times 2^2 = 20 \text{ metres}$$

ii) Range when it hits the ground ($y = 0$)
or twice the time to reach maximum height.
hence when $t = 4$.

$$\therefore R = 20\sqrt{3} \times 4 = 80\sqrt{3} \text{ metres}$$

iii) $x = 20\sqrt{3}t \rightarrow t = \frac{x}{20\sqrt{3}}$

$$\therefore y = 20 \left(\frac{x}{20\sqrt{3}} \right) - 5 \left(\frac{x}{20\sqrt{3}} \right)^2$$

$$\therefore y = \frac{x}{\sqrt{3}} - \frac{x^2}{240}$$

c) i) Using the reference sheet, the equation
of the normal at P is:

$$x + py = ap^3 + 2ap$$

and at N $x = 0$

$$\therefore 0 + py = ap^3 + 2ap \rightarrow y = ap^2 + 2a$$

$$\therefore N(0, ap^2 + 2a)$$

ii) $N(0, ap^2 + 2a); P(2ap, ap^2)$

$$\therefore M(ap, ap^2 + a)$$

$$\text{From } M \quad x = ap \rightarrow p = \frac{x}{a}$$

$$\therefore y = a \left(\frac{x}{a} \right)^2 + a$$

$$\therefore y = \frac{x^2}{a} + a \rightarrow y = \frac{x^2 + a^2}{a}$$

$\therefore x^2 = a(y - a)$ which is a parabola

with focal length $\frac{a}{4}$

d) i) $v^2 = 5 + 4x - x^2$

The particle is at rest when $v^2 = 0$ (ie $v = 0$).

$$\therefore 5 + 4x - x^2 = 0$$

$$\therefore (x - 5)(x + 1) = 0 \rightarrow x = -1 \text{ and } x = 5$$

ii) amplitude = 5 - centre of motion

$$= 5 - \frac{-1 + 5}{2} = 3$$

iii) $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dx} \left(\frac{1}{2} (5 + 4x - x^2) \right)$

$$\therefore \ddot{x} = 2 - x$$

iv) $\ddot{x} = -1^2 (x - 2) \rightarrow n = 1$

$$\therefore P = \frac{2\pi}{n} = 2\pi$$

Question 14.

a) i) $\angle APB = 90^\circ$ (angle in a semi-circle)

$\therefore \angle APQ = 90^\circ$ (angle sum straight line)

Also $\angle AOQ = 90^\circ$ (angle sum straight line)

$\therefore \angle APQ = 90^\circ = \angle AOQ$

hence $AOPQ$ is a cyclic quad ($\angle$'s in same segment)

ii) $\angle AQO = \angle APO$ ($\angle$'s in same segment/cyc quad $AOPQ$)

Also $PSOB$ is a cyclic quad (opp $\angle$'s supplementary)

$\angle APO = \angle OBS$ ($\angle$'s in same segment/cyc quad $PSOB$)

$\therefore \angle AQO = \angle OBS$

b) i) Since the sequence is geometric:

$$\frac{\gamma}{\beta} = \frac{\beta}{\alpha} \rightarrow \beta^2 = \alpha\gamma$$

$$\text{ii) } \alpha\beta\gamma = -\frac{d}{a} = 27$$

$$\text{iii) Using (i) and (ii): } \beta^3 = 27 \rightarrow \beta = 3$$

$$\text{also } \alpha + \beta + \gamma = -\frac{9}{2} \rightarrow \boxed{\alpha + \gamma = -\frac{15}{2}}$$

$$\text{and } \alpha\beta + \alpha\gamma + \beta\gamma = -\frac{27}{2}$$

$$\therefore 3\alpha + \alpha\gamma + 3\gamma = -\frac{27}{2}$$

$$\therefore 3(\alpha + \gamma) + \alpha\gamma = -\frac{27}{2} \rightarrow \boxed{\alpha\gamma = 9}$$

$$\therefore \alpha + \frac{9}{\alpha} = -\frac{15}{2}$$

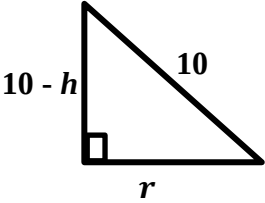
$$\therefore 2\alpha^2 + 15\alpha + 18 = 0$$

$$\therefore (2\alpha + 3)(\alpha + 6) = 0$$

$$\therefore \alpha = -\frac{3}{2} \text{ and } \alpha = -6$$

hence the roots are $-6, -\frac{3}{2}, 3$.

c) i)



$$r^2 + (10 - h)^2 = 10^2$$

$$r^2 = 100 - (10 - h)^2$$

$$= 20h - h^2$$

$$\therefore r = \sqrt{20h - h^2}$$

$$\text{cii) } h = 4, \frac{dV}{dt} = 2, \frac{dh}{dt} = ?$$

$$V = \frac{1}{3}\pi(30h^2 - h^3)$$

$$\therefore \frac{dV}{dh} = \frac{\pi}{3}(60h - 3h^2)$$

$$\therefore \frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$\therefore 2 = \frac{\pi}{3}(60h - 3h^2) \times \frac{dh}{dt}$$

$$\text{and when } h = 4 \rightarrow \frac{dh}{dt} = \frac{1}{32\pi} \text{ cms}^{-1}$$

$$\text{iii) } r = (20h - h^2)^{\frac{1}{2}}$$

$$\therefore \frac{dr}{dh} = \frac{1}{2}(20h - h^2)^{-\frac{1}{2}} \times (20 - 2h)$$

$$= \frac{10 - h}{\sqrt{20h - h^2}}$$

$$\text{Now } \frac{dr}{dt} = \frac{dr}{dh} \times \frac{dh}{dt}$$

$$= \frac{10 - h}{\sqrt{20h - h^2}} \times \frac{1}{32\pi}$$

$$\text{and when } h = 4 \rightarrow \frac{dr}{dt} = \frac{3}{128\pi} \text{ cms}^{-1}$$